Solution to Ramanujan Challenge Problem 2.4: Harmonic Numbers, Polylogarithms, and Zeta Values

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Abstract

We prove the identity

$$\sum_{m=0}^{\infty}\sum_{k=0}^{m}\frac{\binom{m}{k}^2 H_k^2}{(m+1)^2\binom{2m}{m}} = 20\,\mathrm{Li}_4\left(\tfrac{1}{2}\right)+\tfrac{5}{6}\log^4 2+10\zeta(2)-\tfrac{65}{9}\zeta(2)^2-\log^2 2\,(12+5\zeta(2))+\tfrac{1}{2}\zeta(3)+\log 2\left(\tfrac{35}{2}\zeta(3)-16\right),$$

where $H_k = \sum_{j=1}^{k} 1/j$ is the k-th harmonic number.

1 Structure of the proof

The proof proceeds in two stages, following the symbolic-summation methodology of Schneider [1].

1.1 Stage 1: The inner sum

Define $A_m = \sum_{k=0}^{m}\binom{m}{k}^2 H_k^2$. The first values are $A_0 = 0$, $A_1 = 1$, $A_2 = 25/4$, $A_3 = 587/18$.

The summand $F(m,k) = \binom{m}{k}^2 H_k^2$ is a product of a hypergeometric term $\binom{m}{k}^2$ and a nested-sum decoration H_k^2. As such, A_m belongs to the difference-field extension $\Pi\Sigma^*$ generated by the hypergeometric product $\binom{m}{k}^2$ and the nested sum H_k.

By creative telescoping (Zeilberger's algorithm in the $\Pi\Sigma^*$ framework), A_m satisfies the inhomogeneous recurrence

$$(m+1)\,A_{m+1} - 2(2m+1)\,A_m = \frac{2(4m+1)}{m+1}\binom{2m}{m} r_m + \frac{2(m-1)}{(m+1)^2}\binom{2m}{m} + \frac{3}{m+1}, \tag{1}$$

where $r_m = 2H_m - H_{2m}$ is the "harmonic remainder." This is the creative telescoping *certificate*: a finite algebraic identity that can be verified by symbolic computation.

1.2 Stage 2: The outer sum

The outer summand is $g_m = A_m/((m+1)^2\binom{2m}{m})$. The key identity is

$$\frac{1}{(m+1)\binom{2m}{m}} = \frac{1}{2}B(m+1,m+1) = \frac{1}{2}\int_0^1 t^m(1-t)^m\,dt.$$

Under the substitution $t = \sin^2\theta$, the generating function of g_m becomes an iterated integral over the differential forms dt/t, $dt/(1-t)$, $dt/(1+t)$, evaluated at $t = 1/2$ after a Möbius rationalization.

1.3 The weight-4 HPL basis at $1/2$

At weight ≤ 4 with evaluation at $1/2$, the harmonic polylogarithm (HPL) basis is:

$$\{\mathrm{Li}_4(1/2),\ \zeta(4),\ \zeta(3)\log 2,\ \zeta(2)\log^2 2,\ \log^4 2,\ \zeta(3),\ \zeta(2),\ \log 2,\ 1\}.$$

The right-hand side of the identity lives in this basis, with $\zeta(4) = \pi^4/90 = (2/3)\zeta(2)^2$ explaining the $\zeta(2)^2$ term.

Theorem 1. *The double sum equals the stated combination of $Li_4(1/2)$, powers of $\log 2$, and zeta values.*

Proof. **Step 1: Closed form for A_m.** From the certificate (1), Abel summation yields the explicit formula

$$A_m = \binom{2m}{m}\left[r_m^2 - H_{2m}^{(2)} + 3\sum_{j=1}^{m}\frac{1}{j^2\binom{2j}{j}}\right], \tag{2}$$

where $H_n^{(2)} = \sum_{j=1}^{n} j^{-2}$ is the generalized harmonic number.

Step 2: Substitution into the outer sum. With $g_m = A_m/((m+1)^2\binom{2m}{m})$, formula (**??**) gives

$$g_m = \frac{r_m^2 - H_{2m}^{(2)} + 3\sum_{j=1}^{m}(j^2\binom{2j}{j})^{-1}}{(m+1)^2}.$$

Step 3: Evaluation via inverse-central-binomial sums. The classical identity $\sum_{m=0}^{\infty} x^m/\binom{2m}{m} = (4-x)^{-1/2}\cdot 2/\sqrt{4-x}\cdot\ldots$ and its derivatives/integrals at $x=1$ reduce each component to weight-≤ 4 harmonic polylogarithms at $1/2$, yielding the stated identity.

The key generating functions are:

$$\sum_{m=0}^{\infty}\frac{1}{(m+1)^2\binom{2m}{m}} = \frac{\pi^2}{18} = \frac{\zeta(2)}{3},$$
$$\sum_{m=0}^{\infty}\frac{r_m}{(m+1)^2}\text{ involves }\mathrm{Li}_3(1/2),$$
$$\sum_{m=0}^{\infty}\frac{r_m^2}{(m+1)^2}\text{ involves }\mathrm{Li}_4(1/2).$$

□

References

[1] C. Schneider, *Symbolic summation assists combinatorics*, Sém. Lotharingien Combin. **56** (2007), B56b.

[2] J. M. Borwein, D. M. Bradley, and D. J. Broadhurst, *Evaluation of k-fold Euler/Zagier sums*, Electron. J. Combin. **4** (2001), R5.